UML Activity Diagram

Activities and Actions

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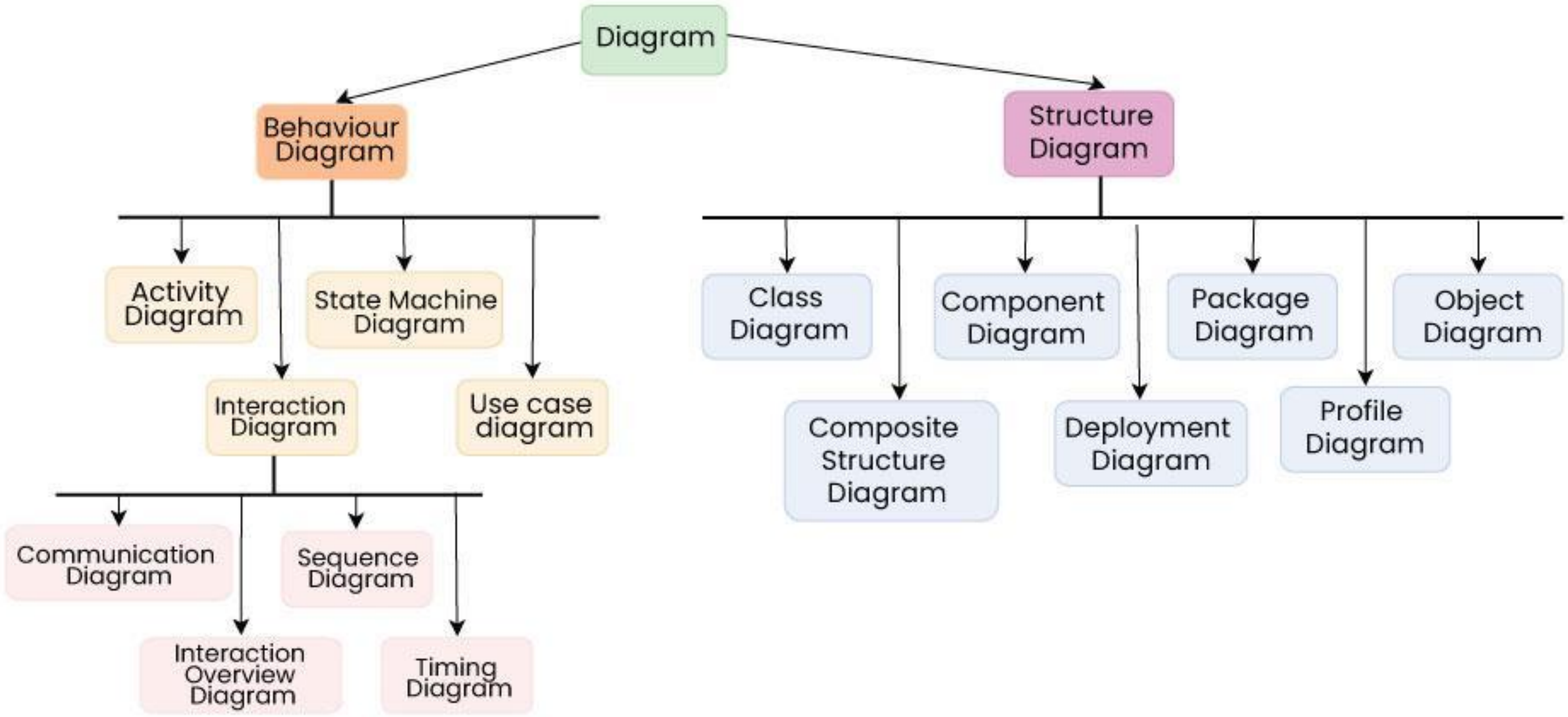
UML

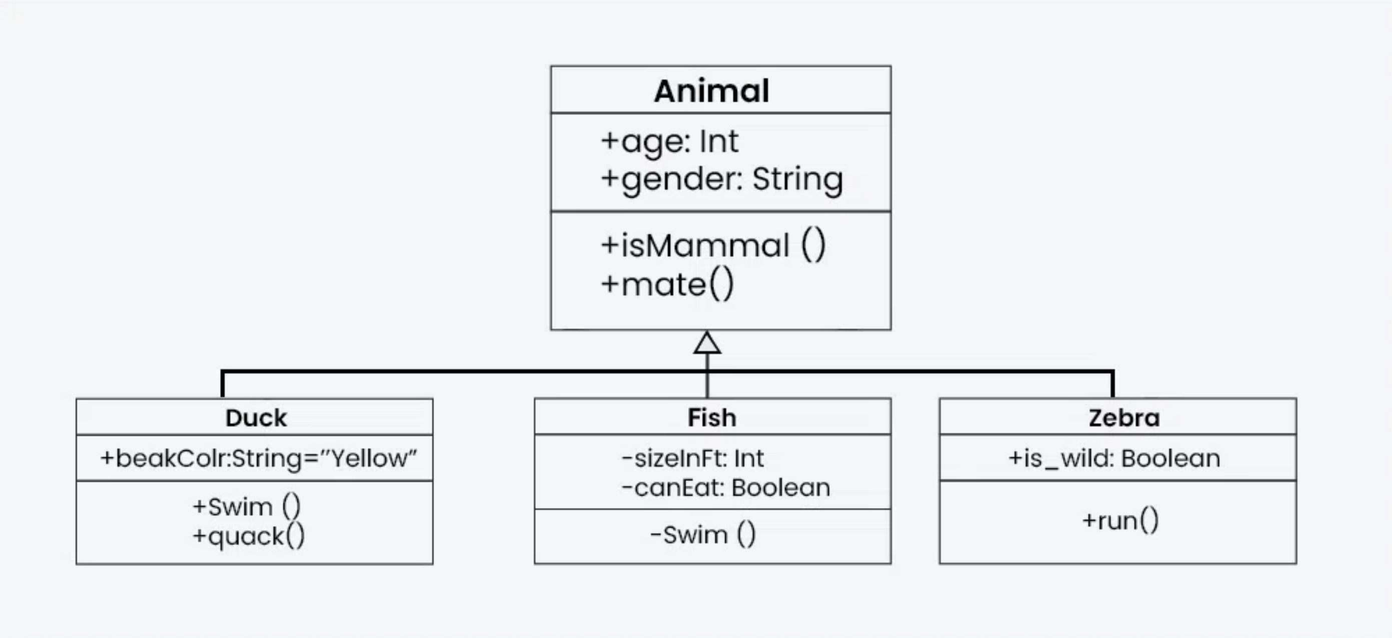
- The Unified Modeling Language (UML) is a standardized, general-purpose visual modeling language used to specify, visualize, construct, and document software systems, business processes, and structural designs



UML Diagrams

Structural diagrams

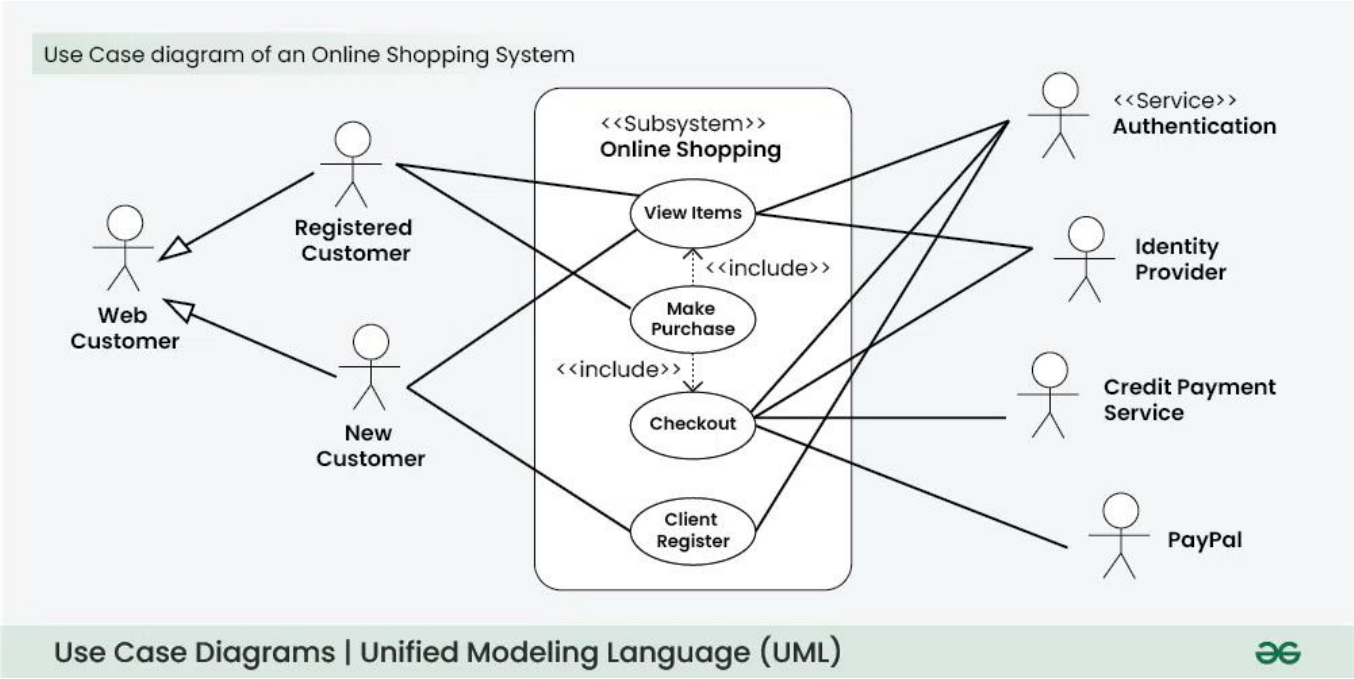
- which map the static, physical, or logical components of a system (e.g., classes, objects, deployment)



Class Diagram example

Behavioral diagrams

- which represent the dynamic, changing interactions, processes, and workflows over time (e.g., activity, sequence, state machine)



Activity Diagram

- Activity diagrams show the steps involved in how a system works, helping us understand the flow of control. They display the order in which activities happen and whether they occur one after the other (sequential) or at the same time
- An activity diagram starts from an initial point and ends at a final point, showing different decision paths along the way.
- UML Activity Diagrams are essentially "advanced flowcharts" that show how a process moves from one step to the next.

Use of Activity Diagram

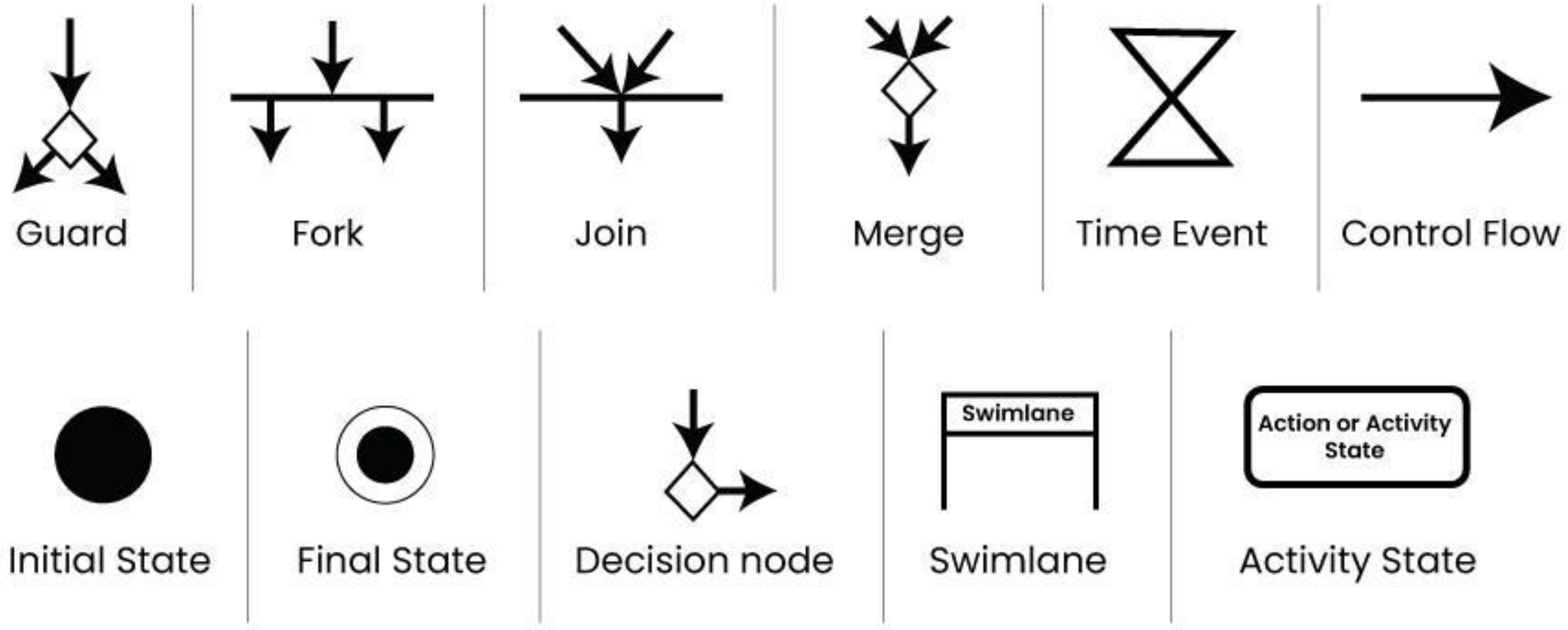
- Activity diagrams are useful in several scenarios, especially when you need to visually represent the flow of processes or behaviors in a system. Here are key situations when you should use an activity diagram:

Use of Activity Diagram

- **Modeling Workflows or Processes**: When you need to map out a business process, workflow, or the steps involved in a use case, activity diagrams help visualize the flow of activities.
- **Concurrent or Parallel Processing**: If your system or process involves activities happening simultaneously, an activity diagram can clearly show the parallel flow of tasks.
- **Understanding the Dynamic Behavior**: When it’s essential to depict how a system changes over time and moves between different states based on events or conditions, activity diagrams are effective.

Use of Activity Diagram

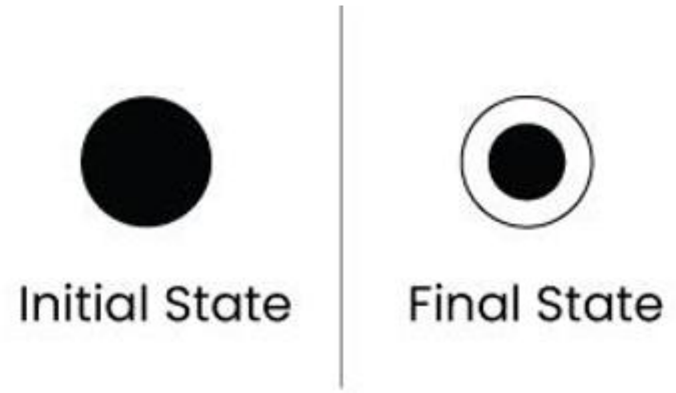
- **Clarifying Complex Logic**: Use an activity diagram to simplify complex decision-making processes with branching paths and different outcomes.
- **System Design and Analysis**: During the design phase of a software system, activity diagrams help developers and stakeholders understand how different parts of the system interact dynamically.
- **Describing Use Cases**: They are useful for illustrating the flow of control within a use case, showing how various components of the system interact during its execution.



Activity Diagram Notations

The Starting and Ending Points

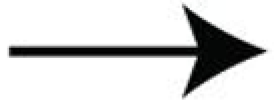
- Initial State: A solid black circle that marks where the process begins.
- Final State: A circle with a dot inside (like a bullseye) that marks the end of the entire process.



Actions and Movement

- Activity State: A rounded rectangle representing a specific task or action being performed (e.g., "Save File").
- Control Flow: An arrow that shows the direction of the process, pointing to what happens next.

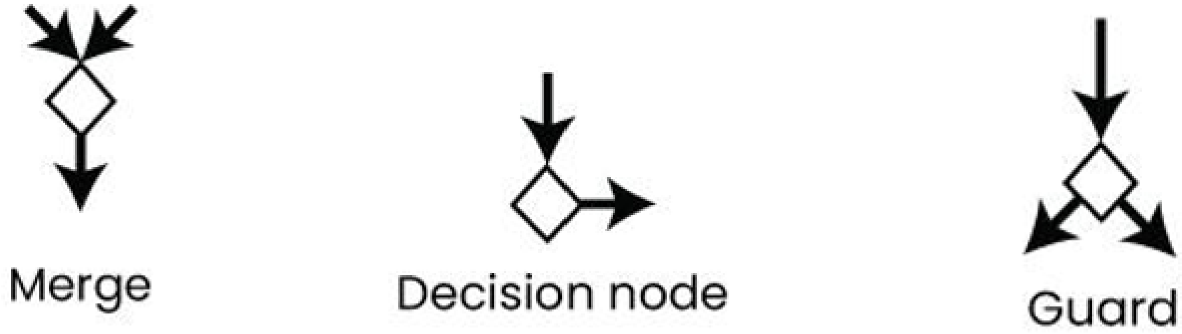
Activity State



Control Flow

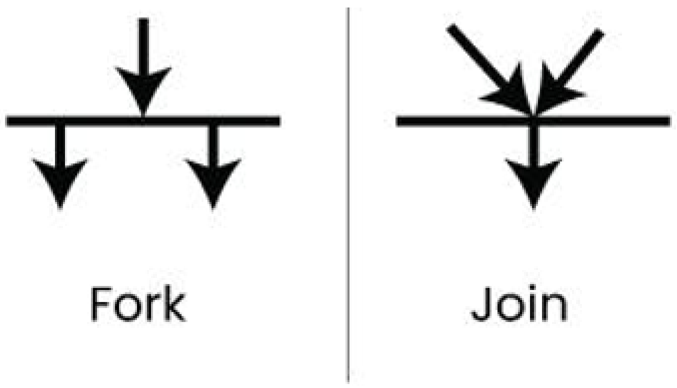
Choosing a Path

- Decision Node: A diamond shape used when the process has to make a choice. It has one entry and multiple exits labeled with conditions (like "Yes" or "No").
- Guard: These are the specific conditions (usually in brackets like [if valid]) written next to arrows leaving a decision node to ensure the flow only goes down the correct path.
- Merge: A diamond used to bring multiple separate paths back into a single flow.



Multitasking (Parallel Processing)

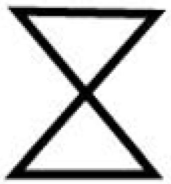
- **Fork**: A thick horizontal or vertical bar used to split one path into two or more tasks that happen at the same time.
- **Join**: A thick bar used to wait for all those parallel tasks to finish before moving on to the next single step.



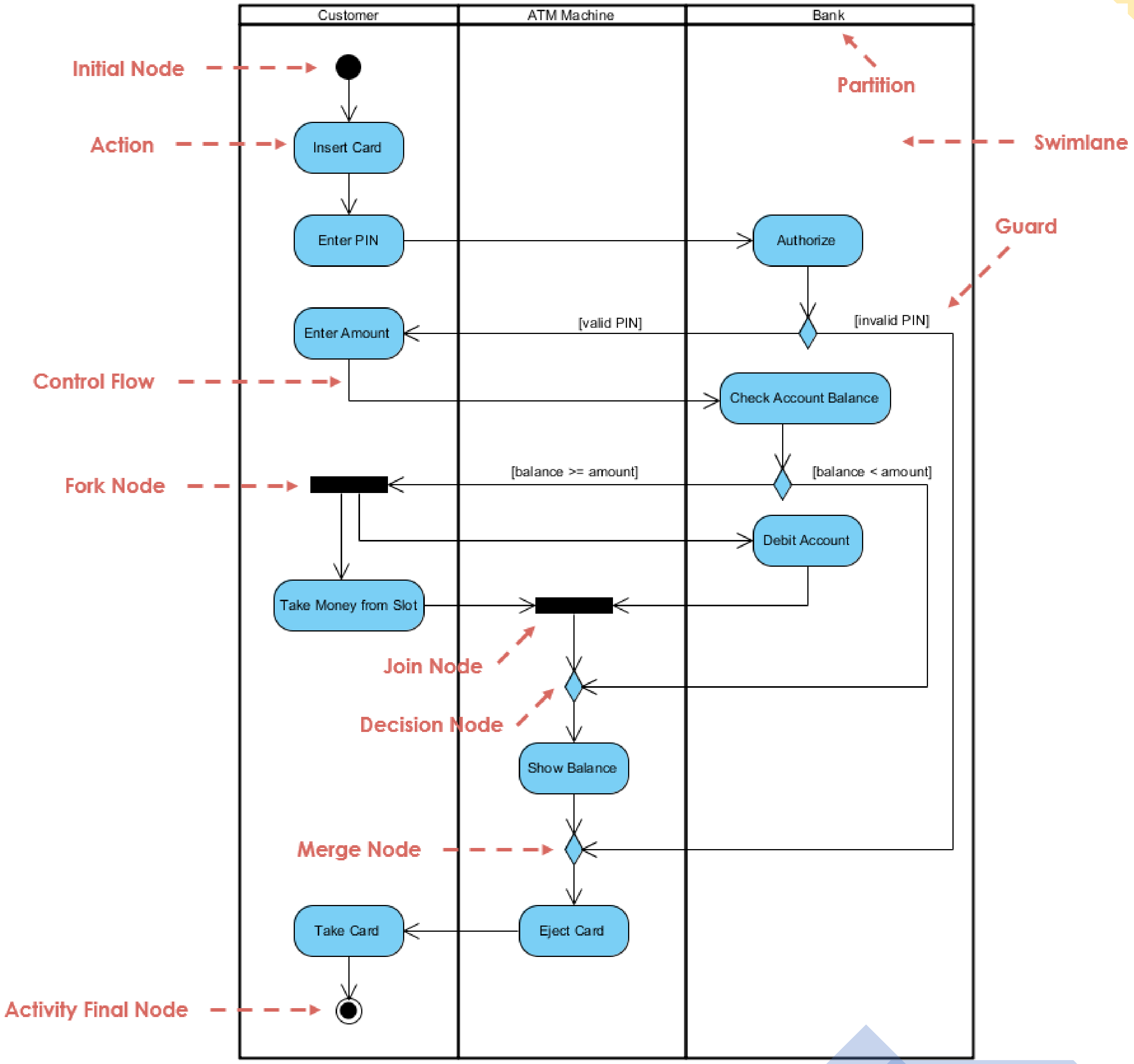
Organization and Timing

- Swimlane: Vertical or horizontal columns used to show who is doing the work (e.g., "Accountant" vs. "Consultant").
- Time Event: Represented by an hourglass shape, this indicates that the process must wait for a specific time or duration to pass before continuing.

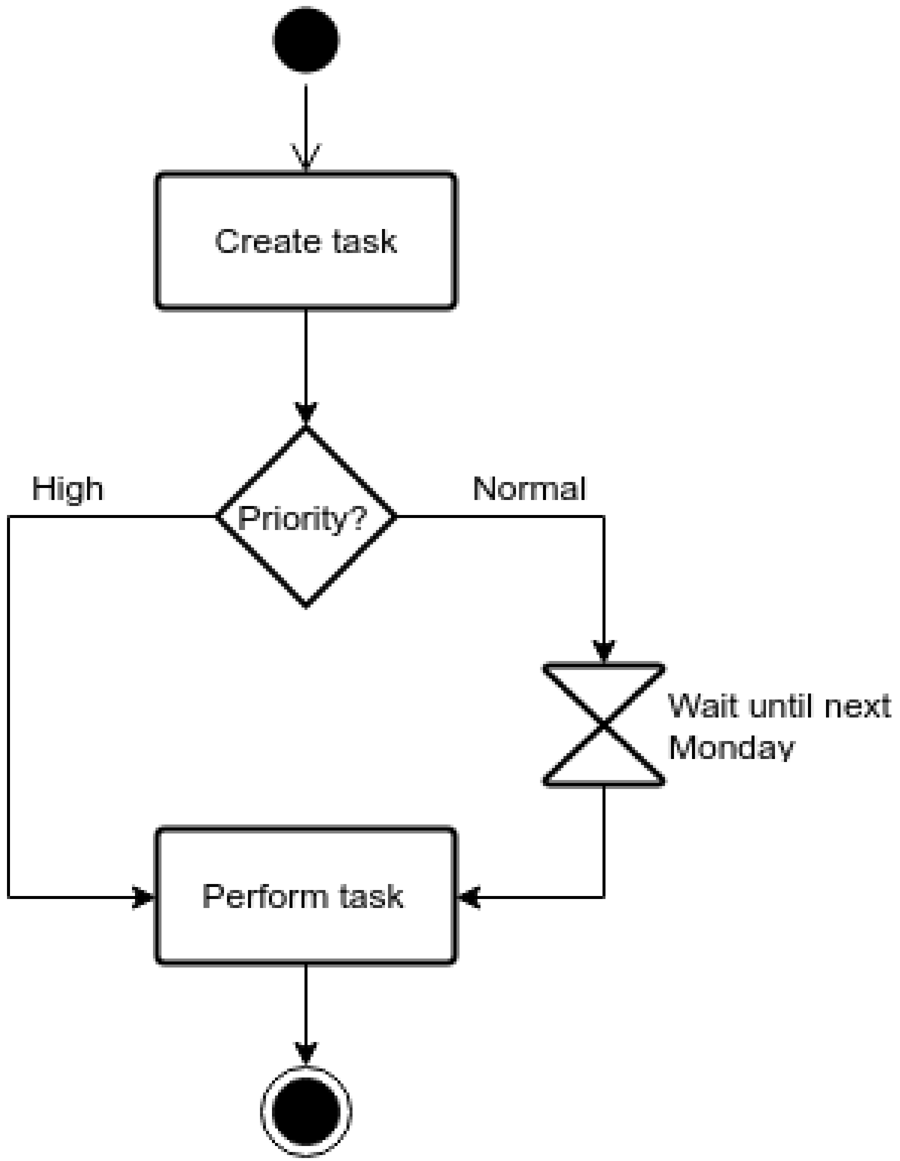
Swimlane

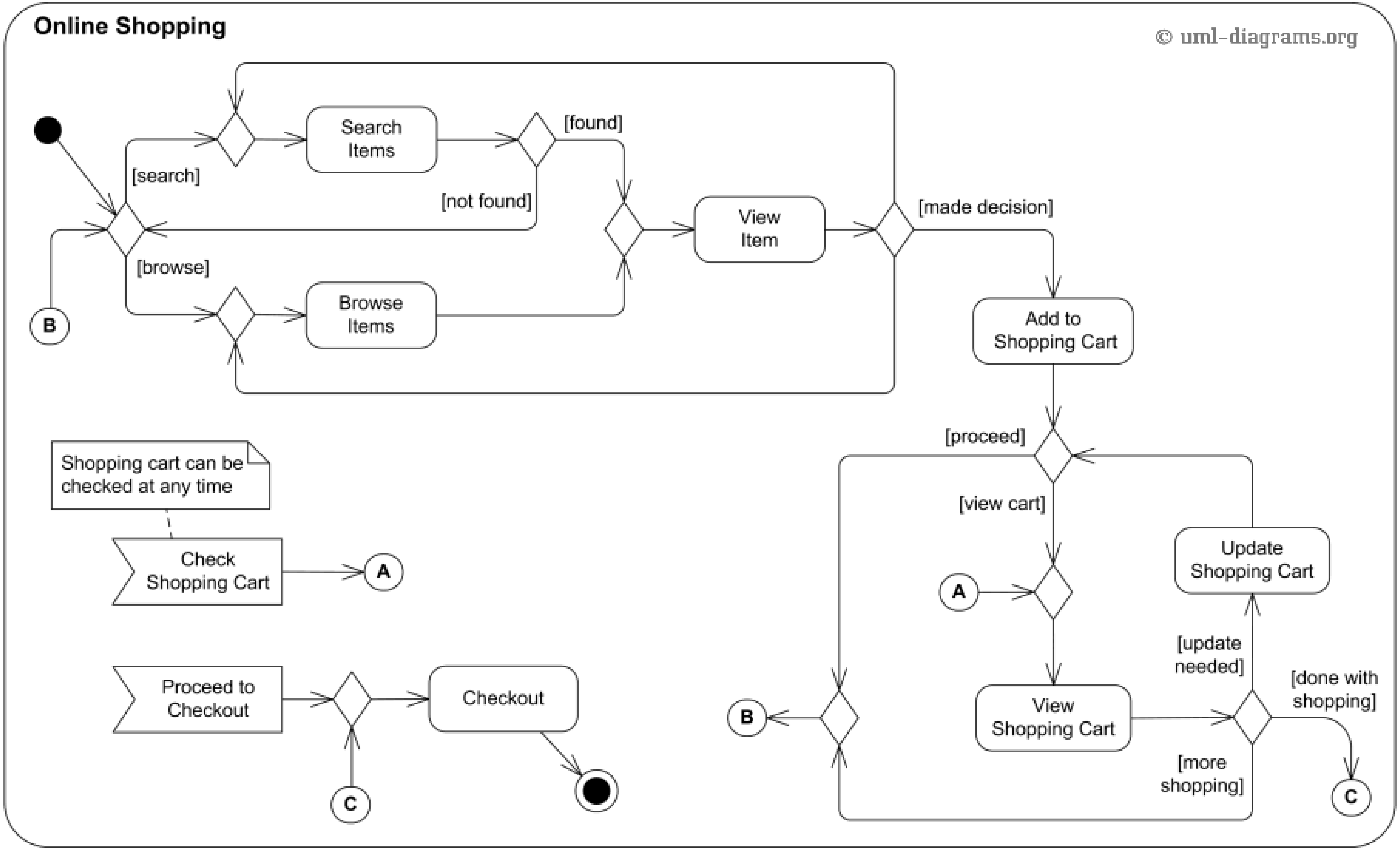


Time Event



Customer
ATM Machine
Bank
Initial Node
Partition
Action
Swimlane
Insert Card
Enter PIN
Authorize
Guard
Enter Amount
[valid PIN]
[invalid PIN]
Control Flow
Check Account Balance
Fork Node
[balance >= amount]
[balance < amount]
Debit Account
Take Money from Slot
Join Node
Decision Node
Show Balance
Merge Node
Take Card
Eject Card
Activity Final Node


Create task
High
Priority?
Normal
Wait until next
Monday
Perform task


Online Shopping
© uml-diagrams.org
Search Items
[found]
[search]
[not found]
View Item
[made decision]
[browse]
Browse Items
Add to Shopping Cart
B
[proceed]
Shopping cart can be checked at any time
[view cart]
Check Shopping Cart
A
Update Shopping Cart
A
[update needed]
[done with shopping]
Proceed to Checkout
Checkout
B
View Shopping Cart
[more shopping]
C
C

Question

- ***Purpose***: *An example of business flow UML activity diagram to process purchase order.*
- ***Summary***: *Requested order is input parameter of the activity. After order is accepted and all required information is filled in, payment is accepted and order is shipped.*
- *User places order system receives it then order is either accepted or rejected if order is processing invoice is sent and the order is shipped user must choose payment method or make payment for simplicity then order is closed*

Practice

- Try to solve this question on your own as a homework
- **The Scenario: Automated Smart Parking System**
- Imagine a smart parking garage that uses an IoT-based system to manage entry. The system must handle vehicle detection, payment verification, and gate control.
- **The Workflow:**
- **Vehicle Entry:** A driver approaches the entrance. A sensor detects the car and triggers the camera to capture the license plate.
- **Validation:** The system checks if the license plate is registered in the database (monthly subscriber) or if it's a guest.
- **Parallel Processing:** * If the user is a subscriber, the system logs the entry time.
    - Simultaneously, the gate motor is activated to open.
- **Guest Flow:** If the user is a guest, the system prompts them to take a printed ticket. The gate only opens *after* the ticket is taken.
- **Finalization:** Once the car passes through a second safety sensor, the gate closes, and the "Available Spaces" display updates.